Hypothesis Testing:- Hypothesis

Null Hypothesis (H0)

- Default Assumption
- (what ever iam trying to prove which is right by assuming it but don't know it is right (or) wrong we should prove statistically.

Eg:- New model has equal accuracy.

Alternative Hypo (H1)

- Testing.
- opposite of Null hypothesis.

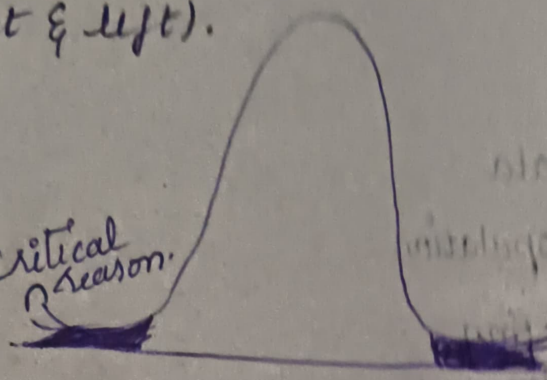
Eg:- New model more accura

⇒ One-Tailed Test :-

⇒ Two-Tailed Test :-

One-Tailed Test :- Critical region on single side.
(either right or left)

Two-Tailed Test :- Critical region on both sides.
(right & left).

=> 

Redcolour - Critical region.
Either right or le

Testing Hypothesis & Workflow :-

1. Define Ho and H1.
↳ State null & alternative hypothesis clearly.

2. Select significance level (α) (Threshold value)
↳ Set threshold (commonly 0.05) for rejecting Ho.

3. choose Test statistic :- z-test, t-test, chi-squared based on data
↳ chi-squared test (categorical data)
↳ z-test } numerical
↳ t-test } data.
↳ sample size is < 30 (low) ⇒ (t-test)
↳ sample size is > 30 (high) ⇒ (z-test)
↳ if we know the population variance (z-test)
⇒ If we don't know " " " (t-test)

4. compute P value (P < α (reject Ho & choose H1 (alterna

5. Rule Decision - ~~Failed to reject (accept it)~~ Reject Ho if P ≤ α; otherwise fail to reject.

6. Interpret Result :- Translate statistical decision into domain context.

Compute P value :-

- If $P < \alpha \Rightarrow$ Reject H_0 & choose H_1 (alternative hypothesis)
  If $P \geq \alpha \Rightarrow$ Reject H_1 & choose H_0 (null hypothesis)

Type1 Error:- If null hypothesis is true but unfortunately we are rejecting it called Type 1 Error.

Type2 Error: If null hypothesis is ~~false~~ but we are accepting it called type1 Error

Assumption:-

Actual	Yes	no
yes	true	mistake
no	mistake	true

Decision:

Actual	Accept null hyp	Reject null hyp
Null Hyp True	True	Type1 Error
Null Hyp False	Type2 Error	True

⇒ **Confidence level:-** Range estimated an unknown population parameter with a stated confidence level $(1-\alpha)$.

when to use which:-

- Use confidence intervals when you need to estimate magnitude & precision.
- Use hypothesis tests when you need to clear yes/no decision about an effect.
  Provide ... & magnitude. the complete